

NT-C-030 | Hardware & Networking | Course Knowledge Profile

Identity, purpose, prerequisites, scope and boundaries

Category	IT Infrastructure
Target learners	Learners targeting desktop, hardware and network-support roles.
Prerequisites	Basic computer use and willingness to perform supervised hardware labs.
Approved duration	16-20 weeks / 180-240 guided hours.
Final outcome	Assemble, configure and troubleshoot computer systems and small networks using safe, documented procedures.
Assessment profile	Infrastructure/Security

Approved tools / platforms

PC hardware tools, Windows/Linux, diagnostic tools, cabling tools, switches/routers and network utilities.

Knowledge boundaries

Include	Quality rule	Exclude / escalate
The eight approved modules, four sections, named tools, projects and mapped entry-level roles.	Require least privilege, authorised labs, configuration evidence, logs, monitoring, rollback and documented troubleshooting.	Never recommend unsafe production testing or treat a scanner output as a verified finding without validation.
Current product/platform procedures only when version and date are known.	Use primary documentation and record the source version.	Do not invent current commands, prices, tax rates, policies or certification status.
Evidence-based career guidance.	Cite tests, projects, capstone, viva and verified resume evidence.	No guaranteed job, placement, salary, interview or employer claim.

LLM answer pattern: identify the course and version, answer from approved records, cite record IDs, distinguish verified facts from guidance, and state any missing evidence.

NT-C-030 | Hardware & Networking | NT-C-030-S01 Deep Knowledge

Modules 1-2 | objectives, concepts, evidence and remediation

Section competency	Components, ESD safety, tools and compatibility; Build, BIOS/UEFI and diagnostics
Completion evidence	Component identification; Working PC build
Section weight	20% of the weighted mock-test average
Assessment	10 MCQ (30 marks) + 4 troubleshooting scenarios (30) + 1 authorised practical lab (40)

Module 1 - Hardware foundation

Objective: Components, ESD safety, tools and compatibility

Observable evidence: Component identification

Concept	Approved knowledge statement	Application behaviour	Evidence
Components	A defined competency within Hardware & Networking. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when components is appropriate; apply it in a Hardware & Networking task; verify the result.	Component identification
ESD safety	A defined competency within Hardware & Networking. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when esd safety is appropriate; apply it in a Hardware & Networking task; verify the result.	Component identification
tools	A defined competency within Hardware & Networking. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when tools is appropriate; apply it in a Hardware & Networking task; verify the result.	Component identification
compatibility	A defined competency within Hardware & Networking. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when compatibility is appropriate; apply it in a Hardware & Networking task; verify the result.	Component identification

Module 2 - Assembly and firmware

Objective: Build, BIOS/UEFI and diagnostics

Observable evidence: Working PC build

Concept	Approved knowledge statement	Application behaviour	Evidence
Build	A defined competency within Hardware & Networking. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when build is appropriate; apply it in a Hardware & Networking task; verify the result.	Working PC build
BIOS/UEFI	A defined competency within Hardware & Networking. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when bios/uefi is appropriate; apply it in a Hardware & Networking task; verify the result.	Working PC build
diagnostics	A defined competency within Hardware & Networking. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when diagnostics is appropriate; apply it in a Hardware & Networking task; verify the result.	Working PC build

Remediation: If the learner cannot explain the purpose, complete the stated task or provide Component identification; Working PC build, return to Modules 1-2, complete one guided practice and then use a new equivalent test form.

NT-C-030 | Hardware & Networking | NT-C-030-S02 Deep Knowledge

Modules 3-4 | objectives, concepts, evidence and remediation

Section competency	Install, drivers, users, updates and recovery; Boot, performance, storage and peripheral faults
Completion evidence	Configured workstation; Fault-resolution log
Section weight	25% of the weighted mock-test average
Assessment	10 MCQ (30 marks) + 4 troubleshooting scenarios (30) + 1 authorised practical lab (40)

Module 3 - Operating systems

Objective: Install, drivers, users, updates and recovery

Observable evidence: Configured workstation

Concept	Approved knowledge statement	Application behaviour	Evidence
Install	A defined competency within Hardware & Networking. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when install is appropriate; apply it in a Hardware & Networking task; verify the result.	Configured workstation
drivers	A defined competency within Hardware & Networking. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when drivers is appropriate; apply it in a Hardware & Networking task; verify the result.	Configured workstation
users	A defined competency within Hardware & Networking. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when users is appropriate; apply it in a Hardware & Networking task; verify the result.	Configured workstation
updates	A defined competency within Hardware & Networking. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when updates is appropriate; apply it in a Hardware & Networking task; verify the result.	Configured workstation
recovery	A defined competency within Hardware & Networking. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when recovery is appropriate; apply it in a Hardware & Networking task; verify the result.	Configured workstation

Module 4 - Troubleshooting

Objective: Boot, performance, storage and peripheral faults

Observable evidence: Fault-resolution log

Concept	Approved knowledge statement	Application behaviour	Evidence
Boot	A defined competency within Hardware & Networking. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when boot is appropriate; apply it in a Hardware & Networking task; verify the result.	Fault-resolution log
performance	A defined competency within Hardware & Networking. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when performance is appropriate; apply it in a Hardware & Networking task; verify the result.	Fault-resolution log
storage	A defined competency within Hardware & Networking. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when storage is appropriate; apply it in a Hardware & Networking task; verify the result.	Fault-resolution log
peripheral faults	A defined competency within Hardware & Networking. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when peripheral faults is appropriate; apply it in a Hardware & Networking task; verify the result.	Fault-resolution log

Remediation: If the learner cannot explain the purpose, complete the stated task or provide Configured workstation; Fault-resolution log, return to Modules 3-4, complete one guided practice and then use a new equivalent test form.

NT-C-030 | Hardware & Networking | NT-C-030-S03 Deep Knowledge

Modules 5-6 | objectives, concepts, evidence and remediation

Section competency	Models, media, Ethernet and devices; IPv4, subnet basics, Wi-Fi and cabling
Completion evidence	Network diagram; Connected lab
Section weight	25% of the weighted mock-test average
Assessment	10 MCQ (30 marks) + 4 troubleshooting scenarios (30) + 1 authorised practical lab (40)

Module 5 - Networking basics

Objective: Models, media, Ethernet and devices

Observable evidence: Network diagram

Concept	Approved knowledge statement	Application behaviour	Evidence
Models	A defined competency within Hardware & Networking. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when models is appropriate; apply it in a Hardware & Networking task; verify the result.	Network diagram
media	A defined competency within Hardware & Networking. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when media is appropriate; apply it in a Hardware & Networking task; verify the result.	Network diagram
Ethernet	A defined competency within Hardware & Networking. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when ethernet is appropriate; apply it in a Hardware & Networking task; verify the result.	Network diagram
devices	A defined competency within Hardware & Networking. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when devices is appropriate; apply it in a Hardware & Networking task; verify the result.	Network diagram

Module 6 - IP and cabling

Objective: IPv4, subnet basics, Wi-Fi and cabling

Observable evidence: Connected lab

Concept	Approved knowledge statement	Application behaviour	Evidence
IPv4	A defined competency within Hardware & Networking. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when ipv4 is appropriate; apply it in a Hardware & Networking task; verify the result.	Connected lab
subnet basics	A logical subdivision of an ip network. In Hardware & Networking, it must be demonstrated through an observable task, not only recalled as a definition.	Explain when subnet basics is appropriate; apply it in a Hardware & Networking task; verify the result.	Connected lab
Wi-Fi	A defined competency within Hardware & Networking. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when wi-fi is appropriate; apply it in a Hardware & Networking task; verify the result.	Connected lab
cabling	A defined competency within Hardware & Networking. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when cabling is appropriate; apply it in a Hardware & Networking task; verify the result.	Connected lab

Remediation: If the learner cannot explain the purpose, complete the stated task or provide Network diagram; Connected lab, return to Modules 5-6, complete one guided practice and then use a new equivalent test form.

NT-C-030 | Hardware & Networking | NT-C-030-S04 Deep Knowledge

Modules 7-8 | objectives, concepts, evidence and remediation

Section competency	Switch/router basics, DHCP, DNS and security; Documentation, backup, ticketing and preventive maintenance
Completion evidence	Small network; Complete support lab
Section weight	30% of the weighted mock-test average
Assessment	10 MCQ (30 marks) + 4 troubleshooting scenarios (30) + 1 authorised practical lab (40)

Module 7 - Network devices

Objective: Switch/router basics, DHCP, DNS and security

Observable evidence: Small network

Concept	Approved knowledge statement	Application behaviour	Evidence
Switch/router basics	A defined competency within Hardware & Networking. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when switch/router basics is appropriate; apply it in a Hardware & Networking task; verify the result.	Small network
DHCP	A defined competency within Hardware & Networking. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when dhcp is appropriate; apply it in a Hardware & Networking task; verify the result.	Small network
DNS	A defined competency within Hardware & Networking. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when dns is appropriate; apply it in a Hardware & Networking task; verify the result.	Small network
security	A defined competency within Hardware & Networking. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when security is appropriate; apply it in a Hardware & Networking task; verify the result.	Small network

Module 8 - Support capstone

Objective: Documentation, backup, ticketing and preventive maintenance

Observable evidence: Complete support lab

Concept	Approved knowledge statement	Application behaviour	Evidence
Documentation	A defined competency within Hardware & Networking. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when documentation is appropriate; apply it in a Hardware & Networking task; verify the result.	Complete support lab
backup	A defined competency within Hardware & Networking. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when backup is appropriate; apply it in a Hardware & Networking task; verify the result.	Complete support lab
ticketing	A defined competency within Hardware & Networking. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when ticketing is appropriate; apply it in a Hardware & Networking task; verify the result.	Complete support lab
preventive maintenance	A defined competency within Hardware & Networking. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when preventive maintenance is appropriate; apply it in a Hardware & Networking task; verify the result.	Complete support lab

Remediation: If the learner cannot explain the purpose, complete the stated task or provide Small network; Complete support lab, return to Modules 7-8, complete one guided practice and then use a new equivalent test form.

NT-C-030 | Hardware & Networking | Skill Ontology & Dependency Map

Atomic skills, learning order and evidence strength

Skill ID	Competency	Depends on	Evidence	Type
NT-C-030-SK01	Hardware foundation	Prerequisite foundation	Component identification	Critical
NT-C-030-SK02	Assembly and firmware	M01 plus earlier foundations	Working PC build	Critical
NT-C-030-SK03	Operating systems	M02 plus earlier foundations	Configured workstation	Supporting
NT-C-030-SK04	Troubleshooting	M03 plus earlier foundations	Fault-resolution log	Supporting
NT-C-030-SK05	Networking basics	M04 plus earlier foundations	Network diagram	Supporting
NT-C-030-SK06	IP and cabling	M05 plus earlier foundations	Connected lab	Critical
NT-C-030-SK07	Network devices	M06 plus earlier foundations	Small network	Supporting
NT-C-030-SK08	Support capstone	M07 plus earlier foundations	Complete support lab	Critical

Evidence-strength hierarchy

Strength	Evidence source	Use
High	Trainer-observed practical, viva, working source/project files and reproducible output.	May support verified mastery when rubric threshold is met.
Medium	Scored mock tests, reviewed assignments and documented portfolio artifacts.	Supports mastery with corroboration.
Low	Resume claim, self-report, attendance or video completion without demonstration.	May trigger verification; cannot alone support Apply Now.

CareerTours must not treat attendance, time spent or content opened as skill mastery.

NT-C-030 | Hardware & Networking | Assessment Knowledge & Sample Items

Non-live examples for LLM training and evaluation

Sample ID	Type	Question	Approved answer / rubric anchor
NT-C-030-S01-EX01	Evidence/application	What evidence best demonstrates the combined competency of Hardware foundation and Assembly and firmware?	The learner should provide Component identification; Working PC build and explain how the work applies Components, ESD safety, tools and compatibility; Build, BIOS/UEFI and diagnostics. Recall alone is insufficient.
NT-C-030-S01-EX02	Scenario/remediation	A learner reports completing Assembly and firmware but cannot produce Working PC build. What should CareerTours do?	Mark the evidence Not Yet Verified, assign practice from NT-C-030-S01, request Working PC build, and allow a new equivalent test only after remediation.
NT-C-030-S02-EX01	Evidence/application	What evidence best demonstrates the combined competency of Operating systems and Troubleshooting?	The learner should provide Configured workstation; Fault-resolution log and explain how the work applies Install, drivers, users, updates and recovery; Boot, performance, storage and peripheral faults. Recall alone is insufficient.
NT-C-030-S02-EX02	Scenario/remediation	A learner reports completing Troubleshooting but cannot produce Fault-resolution log. What should CareerTours do?	Mark the evidence Not Yet Verified, assign practice from NT-C-030-S02, request Fault-resolution log, and allow a new equivalent test only after remediation.
NT-C-030-S03-EX01	Evidence/application	What evidence best demonstrates the combined competency of Networking basics and IP and cabling?	The learner should provide Network diagram; Connected lab and explain how the work applies Models, media, Ethernet and devices; IPv4, subnet basics, Wi-Fi and cabling. Recall alone is insufficient.
NT-C-030-S03-EX02	Scenario/remediation	A learner reports completing IP and cabling but cannot produce Connected lab. What should CareerTours do?	Mark the evidence Not Yet Verified, assign practice from NT-C-030-S03, request Connected lab, and allow a new equivalent test only after remediation.
NT-C-030-S04-EX01	Evidence/application	What evidence best demonstrates the combined competency of Network devices and Support capstone?	The learner should provide Small network; Complete support lab and explain how the work applies Switch/router basics, DHCP, DNS and security; Documentation, backup, ticketing and preventive maintenance. Recall alone is insufficient.
NT-C-030-S04-EX02	Scenario/remediation	A learner reports completing Support capstone but cannot produce Complete support lab. What should CareerTours do?	Mark the evidence Not Yet Verified, assign practice from NT-C-030-S04, request Complete support lab, and allow a new equivalent test only after remediation.

Live bank rule: Generate at least three times the live quantity for each section using the approved Infrastructure/Security blueprint. Human-review every scored item before initial production release.

NT-C-030 | Hardware & Networking | Labs, Projects & Scoring Rubrics

Practical proof of learning and ownership

Level	Approved project	Skills evidenced	Required deliverables
Mini 1	PC assembly	Compatibility, BIOS and OS	Brief; working/source file or reproducible environment; output; validation notes; reflection.
Mini 2	Small-office network	Cabling, IP and Wi-Fi	Brief; working/source file or reproducible environment; output; validation notes; reflection.
Capstone	Support lab	Diagnosis, documentation and recovery	Brief; working/source file or reproducible environment; output; validation notes; reflection.

Standard project rubric

Dimension	Weight	What earns credit
Problem framing	15%	Clear objective, user/business need, assumptions and success criteria.
Technical/design accuracy	25%	Correct methods, appropriate tools and sound decisions.
Completeness	15%	Required features, assets, data or workflow are present.
Testing and validation	15%	Important normal, edge, quality or failure cases are checked.
Evidence and reproducibility	15%	Source/working files, setup, inputs, outputs and version information are available.
Documentation	5%	Concise structure, instructions and limitations.
Viva / ownership	10%	Learner can explain decisions, demonstrate work and respond to a change request.

Require least privilege, authorised labs, configuration evidence, logs, monitoring, rollback and documented troubleshooting.

NT-C-030 | Hardware & Networking | Career Roles, Evidence & Gap Actions

Explainable top-three occupation matching

Rank	Occupation	Default weighted skills	Minimum evidence	Gap action
1	Desktop Support Technician	OS 34% [critical]; hardware 33% [critical]; troubleshooting 33%	Resolved lab tickets	If a critical skill is below 60 or resolved lab tickets is missing, return Improve First or Needs Evidence with the linked module/project.
2	Hardware Technician	Assembly 34% [critical]; diagnostics 33% [critical]; maintenance 33%	PC build evidence	If a critical skill is below 60 or pc build evidence is missing, return Improve First or Needs Evidence with the linked module/project.
3	Network Support Trainee	IP 34% [critical]; devices 33% [critical]; connectivity 33%	Small-network lab	If a critical skill is below 60 or small-network lab is missing, return Improve First or Needs Evidence with the linked module/project.

Course readiness	45% section mocks + 20% mini projects + 25% capstone + 10% viva.
Role readiness	45% role-relevant mastery + 25% projects/capstone + 20% aligned mocks + 10% verified resume evidence.
Evidence confidence	Show separately; Apply Now requires at least 70.
Resume extraction	Use only relevant skills, projects, education, experience and portfolio evidence; minimise personal data before external model processing.

A student may be Apply Now for one mapped occupation and Improve First for another. Return independent evidence and gaps for each role.

NT-C-030 | Hardware & Networking | FAQs, Misconceptions & LLM Response Pack

Course-specific retrieval and response patterns

Approved FAQ	Answer
Who is Hardware & Networking for?	Learners targeting desktop, hardware and network-support roles.
What should I know before starting?	Basic computer use and willingness to perform supervised hardware labs.
How long is the approved learning journey?	16-20 weeks / 180-240 guided hours.
What will I be able to demonstrate?	Assemble, configure and troubleshoot computer systems and small networks using safe, documented procedures.
Does completion guarantee a job?	No. Completion and role readiness are evidence-based guidance; employment, placement, salary and employer selection are not guaranteed.
What if I fail a section?	CareerTours identifies the failed skill tags and module, assigns targeted practice, and permits an equivalent retest after remediation.

Misconception	Approved correction
Attendance proves mastery	False - require tests and observable practical/project evidence.
One total score proves every role fit	False - calculate each occupation using its weighted skills and evidence.
The LLM can add missing details	False - retrieve approved records or state that evidence is insufficient.
Course completion guarantees employment	False - never claim guaranteed job, placement, salary or interview.
Hardware & Networking knowledge never changes	False - version-sensitive tools, laws, taxes, security and platform procedures require review.

Retrieval metadata

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course_code=NT-C-030; corpus_version=3.0;
record_types=course_profile,module,section,skill,assessment_sample,project,occupation_mapping,faq,misconception;
approval_status=approved; effective_date=2026-08-12
```